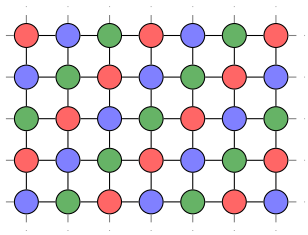
List Colouring the Torus

Odd cycle times odd cycle: where we are, how we got here, and what is left



Prepared for the supervisor meeting, Wednesday 28 October 2026, 4:00 PM

Siddharth Narayan Mishra and Amrutanshu

Every number in this book has a script in the repository `~/work/gt` that reproduces it. Appendix A lists them.

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How to read this book

This book explains one research problem from the start. It assumes you know what a graph, a cycle and a proper colouring are, and nothing else.

- Chapters 1 and 2 explain the problem and what other people already proved.
- Chapter 3 explains the tools, in plain words.
- Chapter 4 is our main finished result: the case where one of the cycles is a triangle.
- Chapters 5 to 7 are about the next case, $C_5 \,\square\, C_n$: what is proved, what is not, and what we tried this morning.
- Chapter 8 is the roadmap with dates. Chapter 9 prepares answers for the meeting.

Coloured boxes tell you the status of every claim:

Proved: example

Green boxes contain things that are fully proved.

Partly done: example

Orange boxes contain things that are proved only in part, or proved with computer help that still needs a human-readable version.

Open: example

Red boxes contain things that are not proved.

For the meeting

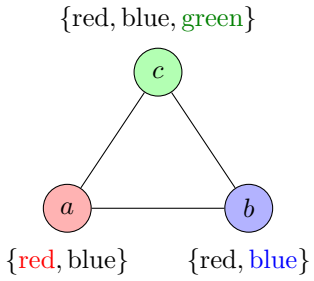
The one-paragraph summary. We want to show that every grid on a torus made of two odd cycles can be coloured from any lists of three colours. The usual polynomial method is known to fail here. We have a complete proof when one cycle is a triangle ($C_3 \,\square\, C_n$, all n). For five-cycles ($C_5 \,\square\, C_n$) we have proved every ingredient except one statement about small two-column pieces, which we call the *window statement*. It is checked by computer for lists of size 3 drawn from a palette of four colours, and survived a targeted search with palettes of five and six colours. This morning we learned why it is hard: lists that are almost constant beat every local cutting argument, so the proof must treat them separately. The general odd case needs the same two ingredients for every odd length, and is still far away.

Chapter 1

The problem

1.1 List colouring

In ordinary colouring, every vertex may use any colour. In *list colouring* each vertex v gets its own list $L(v)$ of allowed colours, and we want a proper colouring where each vertex uses a colour from its own list. A graph is *k-choosable* if this is always possible when every list has k colours, no matter which colours the lists contain. The smallest such k is the *choice number* $\chi_\ell(G)$.

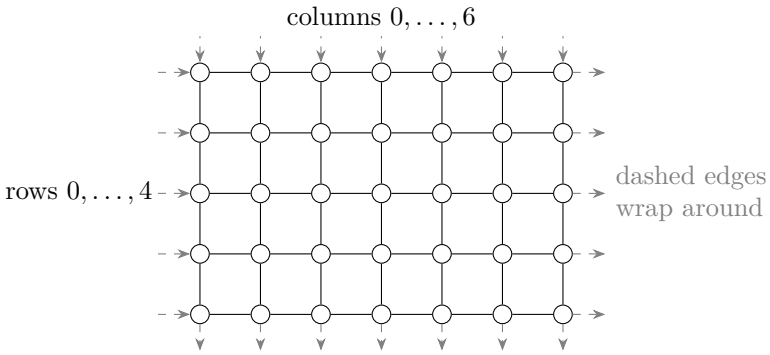


A triangle with lists. The colouring shown uses red, blue, green.

List colouring is harder than ordinary colouring. The adversary picks the lists, and lists can be chosen to block every colouring. A triangle needs 3 colours in any case, so its choice number is at least 3.

1.2 The torus grid

Take a cycle C_m with vertices $0, 1, \ldots, m-1$ and a cycle C_n. The *toroidal grid* $C_m \,\square\, C_n$ has a vertex (i, j) for each *row* i and *column* j. Each vertex is joined to its two neighbours in its column, $(i \pm 1, j)$, and its two neighbours in its row, $(i, j \pm 1)$, where indices wrap around. Picture an $m \times n$ grid whose left edge is glued to its right edge and top edge to its bottom edge: a doughnut. Every vertex has degree 4.



$C_5 \,\square\, C_7$: five rows, seven columns, edges wrap around in both directions.

1.3 The conjecture

Conjecture (Cai, Wang and Zhu, 2010)

Every toroidal grid $C_m \,\square\, C_n$ is 3-choosable.

Since every vertex has degree 4, lists of size 5 are always enough (colour greedily). Lists of size 4 are also known to be enough. The question is whether 3 is enough.

- If m or n is **even**, it is true: Li, Shao, Petrov and Gordeev proved it in 2023.
- If m and n are **both odd**, it is open. This is our problem.

Palette: how to read "with four colours" in this book

In this book every list always has exactly **3** colours. When a computer result says "with a palette of four colours" (or "four-colour windows"), it means the lists are all chosen from a fixed pool of four colours, say $\{0, 1, 2, 3\}$, so each list is one of $\{0, 1, 2\}$, $\{0, 1, 3\}$, $\{0, 2, 3\}$, $\{1, 2, 3\}$. It never means a 4-colouring. A bigger palette gives the adversary more freedom to choose awkward lists. 3-choosability means no palette of any size contains a bad choice of lists, which is why the computer alone cannot settle it.

In plain words

We want to show: if you give every vertex of a doughnut grid (both side lengths odd) any three colours it is allowed to use, you can always colour the grid so that neighbours differ. Nobody knows how to do this yet, and the standard method is proved not to work.

Chapter 2

What was already known

2.1 The polynomial method (Alon and Tarsi)

Give the vertices variables x_v and multiply $(x_u - x_v)$ over all edges. Call this product P. A colouring is proper exactly when $P \neq 0$ at that point. The *Combinatorial Nullstellensatz* says: if some monomial $\prod x_v^{t_v}$ of top degree appears in P with a nonzero coefficient, then the graph can be coloured from any lists with $|L(v)| \geq t_v + 1$.

For a 4-regular graph with N vertices there are $2N$ edges, so we want the monomial $\prod x_v^2$ (every vertex exponent 2, lists of size 3).

Proved: the even case (Li, Shao, Petrov, Gordeev 2023)

If m or n is even, the coefficient of $\prod x_v^2$ in the polynomial of $C_m \,\square\, C_n$ is nonzero, so the torus is 3-choosable. If m and n are both odd, the coefficient is 0, and in fact the *Alon–Tarsi number* is 4: the method can never give 3 for odd tori.

In plain words

There is a famous algebra trick that solves the even case in one stroke. For odd tori the trick provably breaks. So any proof for odd tori must contain at least one step that is not this algebra. In our proofs that step is a clever choice made by looking at the actual lists.

2.2 The smallest case: Galvin

$C_3 \,\square\, C_3$ is the 3×3 "rook's graph": cells of a 3×3 board, adjacent if in the same row or column. It is the line graph of $K_{3,3}$, and Galvin's famous theorem (the solution of the Dinitz problem) shows it is 3-choosable. Galvin used the *kernel method*: orient the edges cleverly so that every induced subgraph has a *kernel* (an independent set that every outside vertex points into). Then lists of size "out-degree +1" are enough.

Proved: $C_3 \,\square\, C_3$ (Galvin 1995)

$C_3 \,\square\, C_3$ is 3-choosable.

That is where the field stood: even tori done, $C_3 \,\square\, C_3$ done, every other odd torus open.

Chapter 3

Our tools in plain words

3.1 Cutting the torus open

The torus is hard because it closes up in both directions. If we delete some columns, what is left is a *strip* $C_m \,\square\, P_\ell$: columns are still cycles, but the rows are now paths. A strip has fewer edges than a torus of the same size, so the algebra has a little room to spare.

Key idea: spare room at the ends

A strip $C_m \,\square\, P_\ell$ has $m\ell$ vertices and $2m\ell - m$ edges, so the algebra has exactly m "units" of spare room. We can spend them at the two end columns:

- 2 units **pin** a vertex: its list shrinks to one colour we choose (shown in red).
- 1 unit **forbids** one colour at a vertex: its list shrinks by one (shown in purple).

Whether a particular way of spending the units works depends on a single number, the *strip coefficient* c_S. If $c_S \neq 0$, it works for every choice of lists.

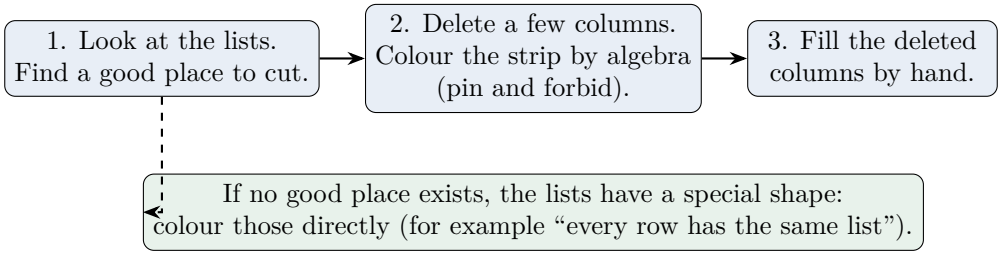
3.2 Computing the strip coefficient

The strip coefficient can be computed column by column: each column contributes a fixed matrix, the *transfer matrix* T, and the coefficient for a strip of length ℓ is roughly "a row vector times $T^{\ell-2}$ times a column vector". For $m = 3$ this matrix is tiny and the coefficient has a closed formula. For $m = 5$ the matrix is 30×30, and we use its eigenvalues: for long strips the biggest eigenvalue dominates, so once its contribution is nonzero the coefficient can never vanish again.

In plain words

We built a calculator for these numbers, proved for $m = 3$ that they are never zero when they should not be, and proved the same for $m = 5$ using the eigenvalues of a 30×30 matrix. The computer does exact arithmetic, so there is no rounding error anywhere.

3.3 The plan of every proof in this book



Step 1 is the non-algebraic step that the polynomial method cannot do on its own.

Chapter 4

The finished case: $C_3 \,\square\, C_n$

Proved: Theorem A

For every $n \geq 3$, $C_3 \,\square\, C_n$ is 3-choosable. So $\chi_\ell(C_3 \,\square\, C_n) = 3$.

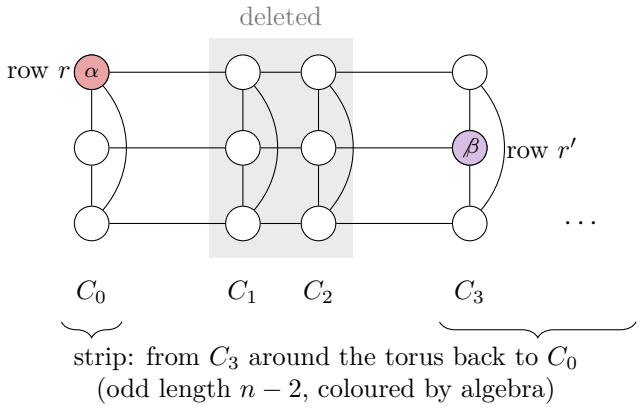
For even n this is the known even case, and $n = 3$ is Galvin. The new part is odd $n \geq 5$. The route is Amrutanshu's; the simplification that made it short is in his work plan; we checked every step, wrote the strip lemma proof ourselves, and verified the finite lemma exhaustively.

4.1 Setting up

Each column of $C_3 \,\square\, C_n$ is a triangle. Let L give every vertex 3 colours. There are two cases, depending on whether the lists change as you walk along a row.

4.2 Case 1: some list changes along a row

Suppose some row r and column j have a colour α that is in $L(r, j)$ but not in $L(r, j+1)$. Call the columns $C_0 = j$, C_1, C_2, C_3.



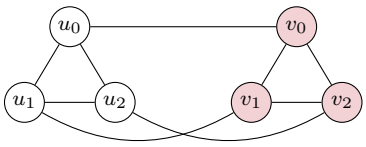
1. Delete C_1 and C_2. The rest is a strip of odd length $n - 2$.
2. Colour the strip by algebra, pinning (r, C_0) to α and forbidding one colour β at (r', C_3), for another row r'. The strip lemma (below) says this works.

3. Fill C_1 and C_2 by hand. Because α is not in $L(r, C_1)$, the vertex (r, C_1) keeps all 3 of its colours; every other vertex of C_1, C_2 keeps at least 2.

The strip lemma for $m = 3$. For odd length ℓ, the coefficient for "pin one vertex at one end, forbid at a different row of the other end" is exactly $\pm 3^{(\ell-1)/2}$, never zero. We proved this by a short exact calculation with cube roots of unity: each column has only 6 possible states, and the sum factorises.

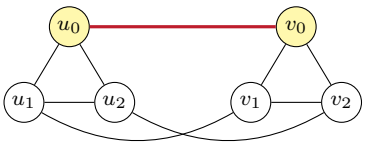
The two-column (prism) lemma. The deleted part is a *prism*: two triangles $u_0u_1u_2$ and $v_0v_1v_2$ joined by three rungs. With 3 colours at u_0 and at least 2 elsewhere, the fill fails only in two patterns:

Pattern Z



all three v-lists are the same 2 colours

Pattern W



u_1, u_2 use up D, v_1, v_2 use up E: both u_0 and v_0 are forced to ζ

Both patterns need the two v-lists in rows r' and r'' to be the *same* two colours. That can only happen in one specific way, and the colour y' that makes it happen can be read off the lists in advance. We choose β to be exactly that colour. Since the strip was coloured with β forbidden at (r', C_3), the bad pattern cannot occur.

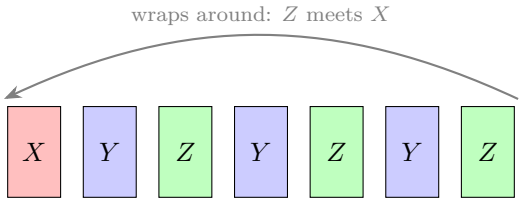
Key idea: why this beats the algebra

β is chosen by looking at the lists. That is the non-algebraic step. The algebra alone (Alon and Tarsi) cannot do this, which is why it fails on odd tori while this proof works.

We verified the two-column lemma by checking every list assignment of the prism up to renaming of colours: 236 million cases, 8,335 of them not colourable, all with the two v-lists equal.

4.3 Case 2: no list changes along any row

Then every row has a constant list A_r. Use Galvin on $C_3 \square C_3$ with these row lists to get three columns X, Y, Z that fit together pairwise. Colour the columns of $C_3 \square C_n$ as $X, Y, Z, Y, Z, \ldots, Y, Z$. Since n is odd, the last column is Z, which fits next to X.



4.4 A second proof for $n = 3k$ (Amrutanshu)

When n is a multiple of 3 (and k odd), there is a completely different proof by the kernel method: colour vertex (i, j) by $(i + j) \bmod 3$, point column edges to the smaller colour and row edges to the larger. Every vertex gets out-degree 2, and a careful argument shows every induced subgraph has a kernel. This gives the stronger property of *3-paintability* (the online version of list colouring). We checked it line by line, and our own counts of kernel-perfect orientations agree with it. His note also reports an exhaustive search showing that for $n = 5, 7, 11, 13$ no orientation of this kind is kernel-perfect, so the kernel method is limited to the family $3 \mid n$.

For the meeting

What to say: "We have a complete proof for $C_3 \,\square\, C_n$ for every n. The polynomial method provably cannot do it; our proof adds one step that looks at the lists to choose a forbidden colour. Every piece is either proved in the write-up or a finite statement we checked exhaustively. A separate kernel-method proof covers $n = 3k$ and even gives paintability."
Still to do before submitting: the hand proof of the two-column lemma (Amrutanshu has one), and a careful literature check that $C_3 \,\square\, C_n$ for odd n is not already in print.

Chapter 5

Moving to five-cycles: what we have

For $C_5 \,\square\, C_n$ each column is a 5-cycle instead of a triangle. We follow the same plan and ask which pieces survive.

Piece	$m = 3$	$m = 5$
Strip lemma (algebra)	proved, short formula	proved, computer-assisted
Every row constant	Galvin	proved (uses Theorem A)
Every column constant	covered by Cases 1 and 2	proved (uses Theorem A)
One free row	not needed	proved by hand (new)
Filling the deleted columns	prism lemma	the window statement

5.1 Rows constant or columns constant

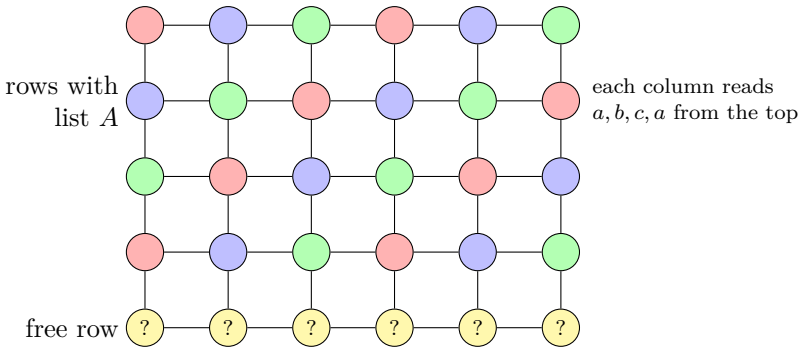
Proved: lists constant along rows or along columns (every odd m, n)

If every row has a constant list, colour $C_m \,\square\, C_3$ with those row lists using Theorem A (it is $C_3 \,\square\, C_m$ turned sideways), take its three columns X, Y, Z, and repeat $X, Y, Z, Y, Z, \ldots$ exactly as before. Columns constant is the same argument turned sideways.

5.2 One free row (new)

Proved: one free row (every odd $m \geq 5$, every n)

If all rows except one have the same constant list A, the torus can be coloured.



How. Colour the rows with list A so that each column reads a, b, c, a (three colours of A in some order), shifting the order by one step from column to column. Then the two neighbours of the free row in each column have the *same* colour, so each vertex of the free row loses at most one colour and keeps at least two. A cycle with lists of size two can be coloured unless all its lists are the same two colours. If that happens, rotate all the colours of A by one step; we showed this always breaks the bad situation.

5.3 The strip lemma for five-cycles

Proved: strip lemma for $m = 5$ (computer-assisted)

For 74 different ways of spending the 5 units of room (one pin plus three forbidden colours, placed at various rows of the two end columns), the strip coefficient is nonzero for *every* odd length.

How. The coefficients for odd lengths satisfy an exact recurrence of order 6, which comes from the minimal polynomial of the 30×30 transfer matrix. For each of the 74 ways the recurrence's largest root dominates from the start, with a nonzero weight, so the coefficient never vanishes. Checked again by exact computation up to length 41.

5.4 Edge deletion and a computer search

Proved: edge deletion for $m = 5, 7$ (computer-assisted)

Remove one edge from $C_5 \,\square\, C_n$ or $C_7 \,\square\, C_n$: the result is 3-choosable (even with one list of size 2). Consequence: in any counterexample, every edge has overlapping lists.

Proved: no small counterexample (certified computation)

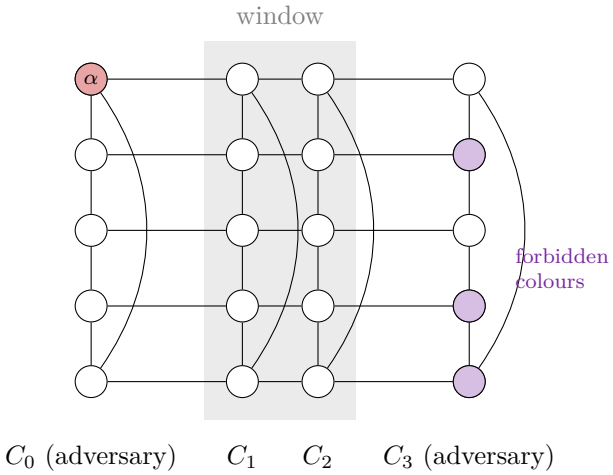
$C_5 \,\square\, C_5$ cannot be made uncolourable by lists of size 3 drawn from 4 colours.

Chapter 6

The missing piece: the window statement

6.1 What a window is

For $m = 5$, a cut deletes two columns C_1, C_2 next to a column C_0 where some row's list changes. The strip colours C_0 and C_3, but the strip colouring is *not under our control* except through the pin and the forbidden colours. So we imagine an *adversary* who colours C_0 and C_3 in the worst possible way.



A window is *good* if some way of spending the 5 units (from the 74 that are proved to work) makes the fill succeed against every adversary. A *cut* is good if its window is good.

The window statement (open)

If every possible cut of a 3-list assignment of $C_5 \,\square\, C_n$ is bad, then the lists are constant along rows, or constant along columns, or have one free row.

If this is true, then $C_5 \,\square\, C_n$ is 3-choosable for every odd n: Chapter 5 handles the three special shapes, and a good cut handles everything else.

6.2 What the computer says

Partly done: the window statement

- **Palette of four colours:** completely true for $n = 5$ and $n = 7$. We checked all 4^{10} windows, found the bad ones, and showed that every assignment in which every cut is bad is row-constant, column-constant or has one free row (600 such assignments for $n = 5$, 2,520 for $n = 7$, all of the last kind).
- **Palettes of five and six colours:** new bad windows do appear (992 found next to the four-colour ones), but they still never combine into anything except the three known shapes.
- **All palettes:** not provable by computer. Already a palette of five colours would need about 370 CPU-hours, and there is no upper limit on the number of colours.

In plain words

The computer agrees with the statement everywhere we can look, but "everywhere we can look" is only small palettes. We need an argument that works for a palette of any size.

6.3 What we learned along the way

- Amrutanshu's suggested way of spending the room (all forbidden colours on one side) is not enough by itself: it fails on 6,904 of the million four-colour windows. Allowing all 74 proved ways, some of which forbid colours on the pin's side too, cuts this to a few hundred.
- A cut can be made in two directions at every place where a list changes. Using both directions removed every remaining problem except the special shapes.
- With five or more colours, a careless model of the pinned colour α created 23,280 fake "new obstructions". Modelling α exactly removed all of them. Every one of those fake cases was also colourable, so there was never a counterexample.

Chapter 7

This morning: attacking the window statement

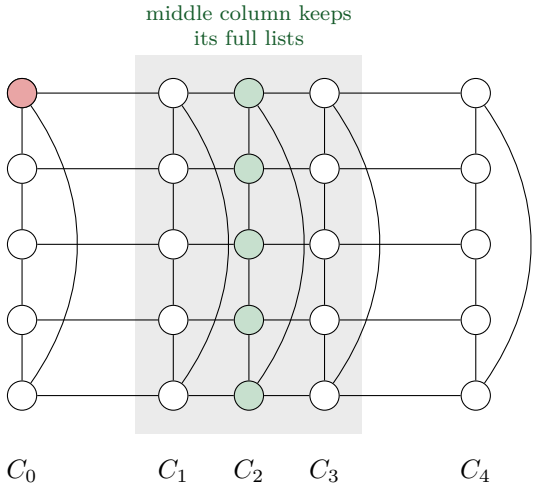
7.1 Attempt 1: a prism lemma for five-cycles

For $m = 3$ the fill failed in only two patterns. We hoped the same for the five-cycle version ($C_5 \,\square\, K_2$, ten vertices). Sampling random lists shows it is **not** so: the failures come in many shapes, including ones where no two lists are equal. A hand classification would be a very long case analysis.

Open: a short characterisation of five-cycle fill failures

Does not exist in the simple form that worked for triangles.

7.2 Attempt 2: cut three columns instead of two



If we delete three columns, the middle one is never touched by the adversary, and we get to choose how to colour the two outer ones so that the middle one works. This is a much more comfortable position. With only the pin (no forbidden colours at all), **none of 400 random four-colour windows and 1 of 400 five-colour-palette windows was bad**, far better than the two-column cut.

7.3 The real obstacle: almost-constant lists

But when the lists are *almost constant* (say every list is $\{0, 1, 2\}$ except one or two), the three-column cut fails too: 75 of 135 windows with one changed list, 2,697 of 8,505 with two. The same was true for the two-column cut. This is not bad luck: when the deleted columns all carry the same three colours, the adversary always has enough control to force a clash.

Key idea: two regimes

No local cutting argument can handle almost-constant lists. A proof must split:

- **Generic lists** (far from constant): cut, preferably three columns, and fill by hand.
- **Almost-constant lists**: colour directly with a global pattern, the way the one-free-row lemma does.

The work is to make "almost constant" precise so that both halves can be proved and they cover everything.

Open: the window statement for all palettes

Not proved. The attempt today narrowed it to the two-regime plan above.

Chapter 8

Late morning: a sharper target

This chapter was added at 9:30 AM on the day of the meeting. It records what the searches found while the earlier chapters were being read.

8.1 Can every cut fail somewhere new?

We searched directly for an assignment of $C_5 \,\square\, C_5$ or $C_5 \,\square\, C_7$ (palettes of five or six colours) in which every cut fails but which is *not* one of the three special shapes. Sixteen independent searches, about two million steps in total, never found one. Every search got stuck with at least two places where a cut still works.

8.2 What does a bad window look like?

We first guessed that a bad window must have at least 7 of its 10 lists equal to one set X ("Conjecture N′"). An unbiased search **disproved** this. Here is one of the three bad windows it found, with $X = \{2, 3, 4\}$:

	C_1	C_2
row 0	$\{0, 1, 4\}$	$\{0, 2, 3\}$
row 1	X	X
row 2	X	X
row 3	X	X
row 4	$\{1, 2, 4\}$	$\{1, 3, 4\}$

a 3×2 block of identical lists

All the bad windows we have, 1,395 of them from every search so far, share one feature:

Partly done: Conjecture N″ (strong evidence, not proved)

If a window is bad, then three consecutive rows carry the *same* 3-set X in *both* deleted columns: a rigid 3×2 block of identical lists. In the data the block never consists of the pinned row together with both of its neighbours.

An unbiased search (nine runs, palettes of five to seven colours) looked for a bad window without such a block and found none; with a palette of five it came close but never got there.

In plain words

A cut only ever fails where the lists contain a small rigid block, three rows tall and two columns wide, in which every list is the same. If we can prove that by hand, the rest of the $m = 5$ problem becomes a question about assignments built from such blocks, which is much smaller and more structured.

8.3 A new theorem we can state today

Proved: Theorem B (computer-assisted)

For every odd n, the torus $C_5 \,\square\, C_n$ can be coloured from any lists of size 3 that are drawn from a palette of four colours.

Proof outline.

1. $n = 3$ is Theorem A. Row-constant, column-constant and one-free-row assignments are handled by the lemmas of Chapter 5 (for every palette).
2. Otherwise some list changes along a row, so there is a cut. All 4^{10} four-colour windows were checked: the bad ones (with the real pinned colour) are exactly 184 windows.
3. Only 64 different columns can sit next to a cut. "Every cut fails" is then a closed walk in a finite graph whose states are three consecutive columns (262,144 states, 270,976 edges). We found every part of the graph that carries a closed walk of odd length (81 parts) and checked that each one consists of a single special shape (column-constant or one free row). Since walks of every length are covered, this settles every odd n at once.
4. So if the assignment is not special, some cut is good, and the strip lemma for all odd lengths plus the fill give a colouring.

The program that judges windows with all 74 shapes was re-checked by an independent Python brute force on 40 windows (20 good, 20 bad): all 40 agree.

In plain words

This is a complete result for infinitely many tori, under one restriction: the lists come from four colours. Removing that restriction is exactly the window statement.

8.4 Rigid rows: the sharpest form of the target

Call row t of a window *rigid* if its two deleted lists are the same set, $L(t, C_1) = L(t, C_2)$. In a rigid row the two vertices have almost no room: one of them is forced to copy the colour the adversary put next to the other. Three rigid rows in a row chain these forced copies around the cycle, and the adversary can make the chain contradict itself.

Partly done: Lemma R (strong evidence, not proved)

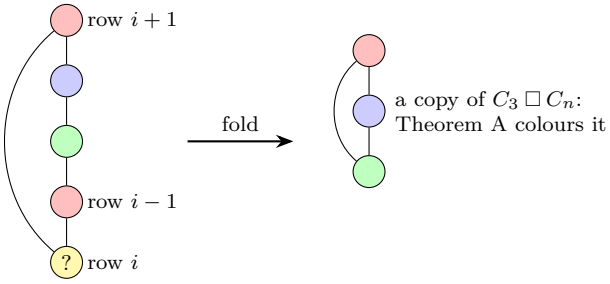
A cut can only fail if at least three rows of its window are rigid.

Evidence: all 4,731 bad (window, α) pairs we know have at least three rigid rows (1,659 have exactly three, 2,852 have four, 220 have five), and sixteen unbiased searches with palettes of five to eight colours found no bad window with two or fewer.

First step towards a proof. The generic case is "no rigid row at all". Amrutanshu's single cut shape, with a natural choice of forbidden colours, already handles 81% of the four-colour windows of this kind; the rest (for example every C_1 list $\{0, 2, 3\}$ and every C_2 list $\{1, 2, 3\}$) need the shapes that also forbid colours on the pinned side. So even the generic case needs a careful, several-page case analysis. That is Phase 1 of the plan.

8.5 A folding trick

Suppose rows $i-1$ and $i+1$ have the same list in every column. Then fold the torus: glue those two rows together. The rows other than i become a copy of $C_3 \,\square\, C_n$ (for $m = 5$), which Theorem A colours. Row i then sees the same colour twice in every column, so each of its vertices loses only one colour, and an odd cycle with lists of size two can almost always be coloured.



Partly done: Folding lemma

This works whenever $C_3 \,\square\, C_n$ stays colourable when one vertex has only two colours ("Lemma S"). Lemma S held in every test (all 393,216 assignments of $C_3 \,\square\, C_3$ with four colours, and 80,000 random ones on larger tori), but is not yet proved: its hard case is again the almost-constant one. The folding lemma would extend the one-free-row lemma and gives a way to reduce m to $m-2$ for lists with this kind of symmetry.

8.6 Revised plan for Phase 1

- Prove Lemma R by hand (a window with at most two rigid rows always has a good cut), starting with windows that have no rigid row.
- Classify the assignments in which every cut fails, using that every column next to a cut contains such blocks. Because the blocks are rigid, this step should reduce to a finite check.
- Prove Lemma S, which gives the folding lemma.

Chapter 9

Roadmap and dates

9.1 Where the whole problem stands

Case	Status	What is missing
m or n even	known (2023)	nothing
$C_3 \,\square\, C_n$	proved (ours)	hand proof of the prism lemma, literature check
$C_5 \,\square\, C_n$	reduced to one statement	the window statement (two regimes)
$C_7 \,\square\, C_n$	some pieces	strip lemma for $m = 7$, window statement
all odd m, n	open	strip lemma for every m, window statement for every m

9.2 The plan

Phase 0, by Fri 30 Oct. Finish the $C_3 \,\square\, C_n$ paper: merge the hand proof of the prism lemma, full literature check, agree authorship.

↓

Phase 1, by Fri 6 Nov. Prove Lemma R (a cut can only fail when three rows of the window are rigid), starting with the case of no rigid row; prove Lemma S (gives the folding lemma).

↓

Phase 2, by Fri 20 Nov. Generic regime: prove that generic windows have a good three-column cut (hand proof with computer-checked case splits), and prove the even-length strip lemma for the shapes used (same method as the odd-length one, about a day).

↓

Phase 3, by Fri 27 Nov. Assemble $C_5 \,\square\, C_n$, verify everything, write it up. **Target for $C_5 \,\square\, C_n$: end of November.**

↓

Phase 4, December to January. General odd m. Needs (a) a strip lemma that works for every m at once (today it is one computation per m), and (b) the two-regime argument written uniformly in m. **Decision point mid December:** if no uniform strip argument is in sight, do $C_7 \,\square\, C_n$ by computer and publish $m = 3, 5, 7$.

In plain words

Honest dates: $C_3 \,\square\, C_n$ is done now (paper by the end of this week). $C_5 \,\square\, C_n$ is realistic by the end of November, with real risk in Phase 2. The full odd case has no promised date: the earliest plausible finish is the end of January 2027, and only if two new uniform arguments are found. Each phase ends with something we can show, even if the next phase stalls.

Chapter 10

Questions to expect, and answers

Is $C_3 \,\square\, C_n$ really new?
Li, Shao, Petrov and Gordeev (2023) state that the odd case is open, and we have found no later paper that closes it. A full check of everything citing their paper and the recent dynamic survey is planned for this week.

Why does your proof work when the Alon–Tarsi method provably fails?
Our proof uses the algebra only on a strip, which has spare room, and then makes one choice by looking at the lists (the forbidden colour β). That choice is not algebraic, so the impossibility result for the torus does not apply.

Why should we trust the computer parts?
All arithmetic is exact (integers and the cube roots of unity, no rounding). Each result was reproduced by at least two independent methods. The finite lemma was checked over every list assignment up to renaming colours. The SAT result comes with a certificate checked by a second solver. Every number has a script in the repository.

What exactly is missing for $C_5 \,\square\, C_n$?
One statement: when every cut fails, the lists have one of three special shapes. Everything else is proved. It is verified for a palette of four colours and survived targeted searches with palettes of five and six. Since this morning we know more: every bad window we have ever found has at least three rigid rows (Lemma R) and a 3×2 block of identical lists (Conjecture N$''$), so the missing proof has a concrete local target.

Why not just compute it?
The palette is unbounded: lists always have 3 colours, but they can be drawn from any number of colours. A palette of five alone would take about 370 CPU-hours, and that would still not cover six or more.

Could the conjecture be false?
No counterexample has appeared: none for $C_5 \,\square\, C_5$ with lists from a palette of four colours (certified), and every candidate obstruction found by the searches was colourable. A counterexample would have to beat every cut, which is very restrictive.

What did you do differently from the original plan?
We proved that the plan's single way of spending the room fails on some windows, found 74 proved alternatives, added cuts in both directions, proved the one-free-row lemma, proved the strip lemma for every odd length when $m = 5$, and found today that almost-constant lists need a separate, global argument.

Who did what?

The $m = 3$ route and the kernel proof are Amrutanshu's; the simplification is in his plan. The verification, the strip lemma proof write-up, the $m = 5$ results and the analysis of the window statement are ours. Credit is shared equally.

Chapter A

Where every number comes from

All paths are inside `~/work/gt/odd-torus/`.

Claim	Script	Result
Strip coefficient, $m = 3$	`ws-W1/strip_coeff.py`, `compare.py`	0 mismatches in 500
Prism lemma, all assignments	`ws-W9/prism_sizes.c`, `prism.c`	0 exceptions
Case 1 extension check	`ws-W9/ext3.c`	0 failures
One-free-row construction	`ws-W9/freerow_check.py`	0 failures in 20,000
Strip lemma $m = 5$, all odd lengths	`ws-W7/prove_strip5.py`	148 cases proved
Edge deletion $m = 5, 7$	`ws-W4/`	see `REPORT.md`
No bad lists from a 4-colour palette on $C_5 \,\square\, C_5$	`ws-W2/certify.py`	UNSAT, certified
Windows, plan shape	`ws-W8/window5.c`	6,904 bad of 1,048,576
Windows, 74 shapes	`ws-W8/window5g.c`	184 bad (exact α)
Every cut bad, $n = 5, 7$	`ws-W8/global_q4_all_n.py`	600 and 2,520, all one free row
Palettes of five and six	`ws-W8/global_alpha.py`	only the known shapes
Five-cycle fill failures	`ws-W10/prism5_sample.py`	many different shapes
Three-column cut	`ws-W10/cut3.py`, `cut3_nearconst.py`	see Chapter 7
Every cut fails, new shapes?	`ws-W11/evade.c`	none found (16 runs)
Conjecture N′	`ws-W11/hunt.c`	false (3 counterexamples)
Conjecture N″	`ws-W11/hunt5.c`	1,395 bad windows agree; hunt found none against
Lemma S evidence	`ws-W11/one_short.py`	0 failures
Lemma R evidence	`ws-W11/huntR.c`	none against (16 runs)
Theorem B, every odd n	`ws-W8/alln_q4.py`	81 odd components, all special
Window verdicts re-checked	`ws-W8/validate_shapes.py`	40 of 40 agree
No rigid row, single shape	`ws-W11/r0_strategy.c`	47,976 of 248,832 need more shapes

Chapter B

Glossary

Choosable G is k-choosable if it can be coloured from any lists of size k.

Torus grid $C_m \,\square\, C_n$ An $m \times n$ grid with both pairs of opposite sides glued.

Strip What is left after deleting some columns: columns are cycles, rows are paths.

Pin Use 2 units of the strip's spare room to force one vertex to a chosen colour.

Forbid Use 1 unit to stop one vertex from using one chosen colour.

Strip coefficient The number that decides whether a pin and forbid pattern works.

Transfer matrix A matrix that computes strip coefficients one column at a time.

Window The deleted columns of a cut, together with the adversary's choices around them.

Kernel An independent set that every vertex outside it points into.

Alon–Tarsi number The best list size the polynomial method can prove. It is 4 for odd tori.

Paintability A harder, online version of choosability.